

Pattern Discovery in Online Retail Transactions

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Abstract—This project proposes an exploratory data mining study on a large-scale online retail transaction dataset. The objective is to discover meaningful patterns in customer purchasing behavior through association rule mining, clustering, and anomaly detection techniques. The analysis focuses on identifying frequent product combinations, natural customer segments, and unusual transactions that deviate from typical purchasing patterns.

I. DATASET DESCRIPTION

This project uses the Online Retail II dataset obtained from Kaggle, which contains transactional data from a UK-based online retail company operating between 2009 and 2011. The dataset consists of over one million records, where each row represents an individual item purchased within a transaction.

The dataset includes eight primary attributes: invoice number, product identifier, product description, quantity purchased, invoice date, unit price, customer identifier, and country. These attributes allow analysis at both the transaction and customer levels.

Several data quality challenges are present, including missing customer identifiers, canceled transactions indicated by negative quantities or special invoice numbers, duplicate entries, and zero-priced items. These issues require preprocessing and cleaning prior to analysis.

II. DISCOVERY QUESTIONS

A. Product Association Patterns

What products are frequently purchased together in the Online Retail II dataset, and how do these associations vary across different countries or time periods? Understanding frequent product combinations can reveal hidden purchasing behaviors and meaningful relationships between items that are not apparent from individual transactions.

B. Customer Segmentation

Are there natural clusters of customers based on their purchasing behavior, such as purchase frequency, spending patterns, and product diversity? Discovering these clusters can help identify distinct customer groups and uncover behavioral patterns without relying on predefined labels.

C. Anomalous Transactions

Which transactions in the dataset appear anomalous in terms of quantity, frequency, or monetary value, and what characteristics distinguish these transactions from typical purchasing behavior? Identifying anomalous transactions can provide insight into unusual purchasing patterns and potential data quality issues.

III. PLANNED TECHNIQUES

A. Association Rule Mining

Association rule mining techniques such as Apriori or FP-Growth will be used to identify frequent itemsets and co-purchase patterns within the transactional data. Metrics including support, confidence, and lift will be used to evaluate the strength of discovered associations. This technique directly addresses the discovery of frequently purchased product combinations.

B. Customer Clustering

Clustering techniques, including K-Means and hierarchical clustering, will be applied to customer-level features derived from transactional data. These features include purchase frequency, total spending, average basket size, and product diversity. Clustering enables the discovery of natural customer segments without predefined labels.

C. Anomaly Detection

Anomaly detection methods will be employed to identify transactions or customers that significantly deviate from typical purchasing behavior. Statistical thresholds and density-based approaches will be used to detect extreme values in quantity, spending, or transaction frequency.

D. Planned Analysis Pipeline

The analysis will follow a structured pipeline consisting of data inspection, data cleaning, feature engineering, association rule mining, customer clustering, anomaly detection, and visualization of discovered patterns.

IV. PRELIMINARY TIMELINE

A. Milestone 2: Data Understanding and Exploration

Milestone 2 will focus on data cleaning and exploratory analysis. Tasks include handling missing values, removing canceled transactions, correcting invalid entries, performing exploratory data analysis, and engineering transaction-level and customer-level features.

B. Milestone 3: Pattern Discovery and Analysis

During Milestone 3, association rule mining, customer clustering, and anomaly detection techniques will be applied. Discovered patterns will be evaluated, refined, and interpreted to ensure meaningful insights.

C. Milestone 4: Final Results and Presentation

Milestone 4 will involve finalizing results, creating visualizations, and preparing the final report and presentation. Findings, limitations, and potential future extensions will be discussed.

D. Anticipated Challenges

Anticipated challenges include handling missing or inconsistent customer identifiers, managing the large dataset size, selecting appropriate parameters for data mining techniques, and distinguishing meaningful anomalies from legitimate but uncommon purchasing behavior.